



Lecture 4: Discrete random variables

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1. Review
2. Random variables
3. Expectation and variance
4. Recitation
5. Summary



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- Independence and conditional independence of two events

$$\mathbf{P}(A \cap B) = \mathbf{P}(A) \cdot \mathbf{P}(B)$$

$$\mathbf{P}(A \cap B \mid C) = \mathbf{P}(A \mid C) \cdot \mathbf{P}(B \mid C)$$

- Independence of a collection of events

$$\mathbf{P}(A_i \cap A_j \cap \cdots \cap A_q) = \mathbf{P}(A_i) \mathbf{P}(A_j) \cdots \mathbf{P}(A_q)$$

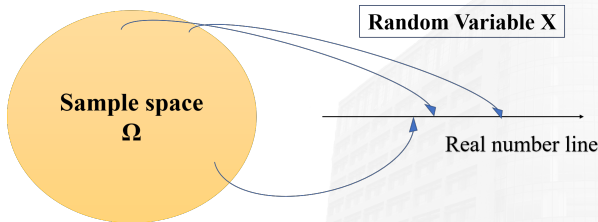
- Counting and binomial probabilities

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

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- An assignment of a value (number) to every possible outcome
- Mathematically: A **function** from the sample space Ω to the real numbers
 - Discrete or continuous values
 - **Discrete RV**: range is either finite or countably infinite



- An assignment of a value (number) to every possible outcome
- Mathematically: A **function** from the sample space Ω to the real numbers
 - Discrete or continuous values
 - **Discrete RV**: range is either finite or countably infinite
- Can have several random variables defined on the same sample space ?
- **Notation:**
 - random variable X
 - numerical value x

- In an experiment with a sequence of 5 tosses of a coin:
 - The number of heads in the sequence [Discrete]
 - But the 5-long sequence is not a RV since it is not a numerical value.
- In an experiment with two rolls of a die:
 - The sum of two rolls [Discrete]
 - The number of sixes in the two rolls [Discrete]
 - The second roll raised to the fifth power [Discrete]
- In an experiment involving the transmission of a message:
 - The number of symbols received in error [Discrete]
 - The delay with which the message is received [Continuous]
 - The time needed to transmit the message [Continuous]

Probability mass function (PMF)

- “probability law” or “probability distribution” of **discrete RV** X

$$\begin{aligned} p_X(x) &= \mathbf{P}(X = x) \\ &= \mathbf{P}(\{\omega \in \Omega \text{ s.t. } X(\omega) = x\}) \end{aligned}$$

- simpler notation is often used, but need to understand the concept
- $p_X(x) \geq 0$, $\sum_x p_X(x) = 1$ due to probability axioms

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- Example: Bernoulli RV**

- One coin toss: $\mathbf{P}(H) = p$ and $\mathbf{P}(T) = 1 - p$
- Let RV $X = 1$ if a head, and $X = 0$ if a tail, and then

$$p_X(k) = \begin{cases} p, & \text{if } k = 1 \\ 1 - p, & \text{if } k = 0 \end{cases}$$

How to compute a PMF $p_X(x)$

- collect all possible outcomes for which X is equal to x
- add their probabilities
- repeat for all x
- **Example:** Two independent rolls of a fair tetrahedral die
 - F : result of first throw
 - S : result of second throw
 - $X = \min(F, S)$

S = Second roll

4				
3				
2				
1				
	1	2	3	4

F = First roll

– $p_X(2) =$

- X : number of heads in n independent coin tosses
- $\mathbf{P}(H) = p$
- Let $n = 4$

$$\begin{aligned} p_X(2) &= \mathbf{P}(HHTT) + \mathbf{P(HTHT)} + \mathbf{P(HTTH)} \\ &\quad + \mathbf{P(THHT)} + \mathbf{P(THTH)} + \mathbf{P(TTHH)} \\ &= 6p^2(1-p)^2 \\ &= \binom{4}{2} p^2(1-p)^2 \end{aligned}$$

- General case

$$p_X(k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \dots, n$$

- X = number of independent coin tosses until first head
- $\mathbf{P}(H) = p$
- Let $n = 4$

$$\begin{aligned} p_X(2) &= \mathbf{P}(X = 2) \\ &= \mathbf{P}(TH) \\ &= (1 - p)p \end{aligned}$$

- General case

$$p_X(k) = \mathbf{P}(TT \cdots TH) = (1 - p)^{k-1}p, \quad k = 1, 2, \dots$$

- PMF of a Poisson RV

$$p_X(k) = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

where λ is a positive parameter

- This is a legitimate PMF because

$$\sum_{k=0}^{\infty} e^{-\lambda} \frac{\lambda^k}{k!} = e^{-\lambda} \left(1 + \lambda + \frac{\lambda^2}{2!} + \frac{\lambda^3}{3!} + \dots \right) = e^{-\lambda} e^{\lambda} = 1$$

- A good approximation for binomial PMF for large n and small p :
let $\lambda = np$ (hint: plug in $p = \lambda/n$ in the RHS, and let $n \rightarrow \infty$)

$$e^{-\lambda} \frac{\lambda^k}{k!} \approx \frac{n!}{k!(n-k)!} p^k (1-p)^{n-k} \quad k = 0, 1, \dots, n$$

- Given a RV X , one can generate other RV Y by applying various transformations on X , i.e., $Y = g(X)$
 - linear function: $Y = aX + b$
 - log function: $Y = \log X$
- If X is discrete with PMF p_X , then Y is also discrete, with PMF p_Y given by

$$p_Y(y) = \sum_{\{x|g(x)=y\}} p_X(x)$$

- Example:** If X is uniform on $\{0, \pm 1, \pm 2\}$, then $Y = |X|$ with PMF

$$p_Y(y) = \begin{cases} 2/5, & y = 1, 2 \\ 1/5, & y = 0 \\ 0, & \text{otherwise} \end{cases}$$

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Definition

The **expected value** (also called the **expectation** or the **mean**) of a random variable X , with PMF p_X , is defined as

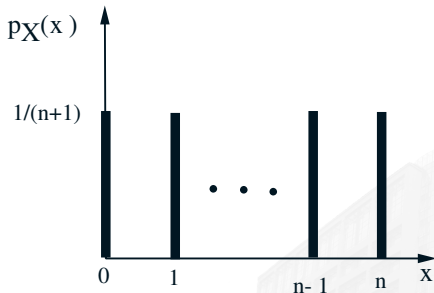
$$\mathbf{E}[X] = \sum_x x p_X(x)$$

• Remarks:

- PMF gives all information, but sometimes need to summarize into a single value
- The “Center of gravity” of PMF
- Average in large number of repetitions of the experiment (to be substantiated later)
- Implicitly assumes the expectation is well-defined (the sum converges for the countably finite case)

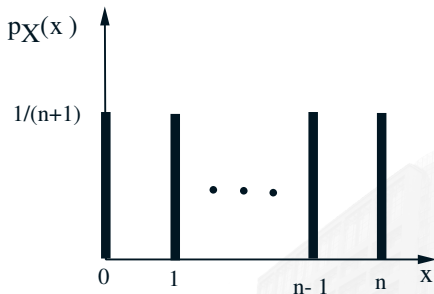
Example

- X is uniform on $0, 1, \dots, n$. $\mathbf{E}[X] = ?$



Example

- X is uniform on $0, 1, \dots, n$. $\mathbf{E}[X] = ?$



$$\mathbf{E}[X] = 0 \cdot \frac{1}{n+1} + 1 \cdot \frac{1}{n+1} + \cdots + n \cdot \frac{1}{n+1} =$$

Properties of expectations

- Let X be a RV and let $Y = g(X)$
 - Hard: $\mathbf{E}[Y] = \sum_y y p_Y(y)$
 - Easy: $\mathbf{E}[Y] = \sum_x g(x) p_X(x)$



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 - Hard: $\mathbf{E}[Y] = \sum_y y p_Y(y)$
 - Easy: $\mathbf{E}[Y] = \sum_x g(x) p_X(x)$
- Proof:

$$\begin{aligned}\mathbf{E}[Y] &= \sum_y y p_Y(y) \\ &= \sum_y y \sum_{\{x|g(x)=y\}} p_X(x) \\ &= \sum_y \sum_{\{x|g(x)=y\}} g(x) p_X(x) \\ &= \sum_x g(x) p_X(x).\end{aligned}$$

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- Caution: In general, $\mathbf{E}[g(X)] \neq g(\mathbf{E}[X])$
- **Properties of Expectation:** If α, β are constants, then
 - $\mathbf{E}[\alpha] =$
 - $\mathbf{E}[\alpha X] =$
 - $\mathbf{E}[\alpha X + \beta] =$
- Expectation of linear functions of a RV

Definition

The **variance** $\text{var}(X)$ of a RV X is defined by

$$\text{var}(X) = \mathbf{E} \left[(X - \mathbf{E}[X])^2 \right]$$

- **Recall:** $\mathbf{E}[g(X)] = \sum_x g(x)p_X(x)$
- Calculation

$$\begin{aligned} \text{var}(X) &= \mathbf{E} \left[(X - \mathbf{E}[X])^2 \right] \\ &= \sum_x (x - \mathbf{E}[X])^2 p_X(x) \end{aligned}$$

- Standard deviation (same unit): $\sigma_X = \sqrt{\text{var}(X)}$

- **Second moment:** $\mathbf{E}[X^2] = \sum_x x^2 p_X(x)$
- **Calculation of Variance**

$$\begin{aligned}\text{var}(X) &= \sum_x (x - \mathbf{E}[X])^2 p_X(x) \\ &= \mathbf{E}[X^2] - (\mathbf{E}[X])^2\end{aligned}$$

Variance in terms of moments expression

$$\text{var}(X) = \mathbf{E}[X^2] - (\mathbf{E}[X])^2$$

Properties of Variance:

- $\text{var}(X) \geq 0$
- $\text{var}(\alpha X + \beta) = \alpha^2 \text{var}(X)$

- Consider the random variable X , which has the PMF

$$p_X(x) = \begin{cases} 1/9, & \text{if } x \text{ is an integer in the range } [-4, 4], \\ 0, & \text{otherwise.} \end{cases}$$

Solution:

- $\mathbf{E}[X] = 0$



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Solution:

- $\mathbf{E}[X] = 0$
- Let $Z = (X - \mathbf{E}[X])^2$

$$p_Z(z) = \begin{cases} 2/9, & \text{if } z = 1, 4, 9, 16, \\ 1/9, & \text{if } z = 0, \\ 0, & \text{otherwise.} \end{cases}$$

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- $\text{var}(X) = \sum_z zp_Z(z) = 0 \cdot \frac{1}{9} + 1 \cdot \frac{2}{9} + 4 \cdot \frac{2}{9} + 9 \cdot \frac{2}{9} + 16 \cdot \frac{2}{9} = \frac{60}{9}$

- Bernoulli RV X with PMF

$$p_X(k) = \begin{cases} p, & \text{if } k = 1 \\ 1 - p, & \text{if } k = 0 \end{cases}$$

- Mean of RV X

$$\mathbf{E}[X] = 1 \cdot p + 0 \cdot (1 - p) = p$$

- Second moment of RV X

$$\mathbf{E}[X^2] = 1^2 \cdot p + 0 \cdot (1 - p) = p$$

- Variance of RV X

$$\text{var}(X) = \mathbf{E}[X^2] - (\mathbf{E}[X])^2 = p - p^2 = p(1 - p)$$

- **Discrete Uniform RV** X with PMF

$$p_X(k) = \begin{cases} \frac{1}{b-a+1}, & \text{if } k = a, a+1, \dots, b \\ 0, & \text{otherwise} \end{cases}$$

- Mean of **RV** X

$$\mathbf{E}[X] = \frac{a+b}{2}$$

- **Variance of RV** X

$$\text{var}(X) = \frac{(b-a+1)^2 - 1}{12} = \frac{(b-a)(b-a+2)}{12}$$

- **Poisson RV** X with PMF

$$p_X(k) = e^{-\lambda} \frac{\lambda^k}{k!}, \quad k = 0, 1, 2, \dots$$

- Mean of **RV** X

$$\begin{aligned} \mathbf{E}[X] &= \sum_{k=0}^{\infty} k e^{-\lambda} \frac{\lambda^k}{k!} = \sum_{k=1}^{\infty} k e^{-\lambda} \frac{\lambda^k}{k!} \quad (\text{the } k=0 \text{ term is } 0) \\ &= \lambda \sum_{k=1}^{\infty} e^{-\lambda} \frac{\lambda^{k-1}}{(k-1)!} \\ &= \lambda \sum_{m=0}^{\infty} e^{-\lambda} \frac{\lambda^m}{m!} \quad (\text{let } m = k-1) \\ &= \lambda \end{aligned}$$

- **Variance of RV:** $\text{var}(X) = \lambda$

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1. Some proofs.

(a) Derive the expected value rule for functions of random variables

$$\mathbf{E}[g(X)] = \sum_x g(x)p_X(x)$$

(b) Derive the property for the mean and variance of a linear function of a random variable $Y = aX + b$

$$\mathbf{E}[Y] = a\mathbf{E}[X] + b, \quad \text{var}(Y) = a^2\text{var}(X)$$

(c) Derive $\text{var}(X) = \mathbf{E}[X^2] - (\mathbf{E}[X])^2$

2. A marksman takes 10 shots at a target and has probability 0.2 of hitting the target with each shot, independently of all other shots. Let X be the number of hits.

- (a) Calculate and sketch the PMF of X .
- (b) What is the probability of scoring no hits?
- (c) What is the probability of scoring more hits than misses?

2. A marksman takes 10 shots at a target and has probability 0.2 of hitting the target with each shot, independently of all other shots. Let X be the number of hits.

- (d) Find the expectation and the variance of X .
- (e) Suppose the marksman has to pay \$3 to enter the shooting range and he gets \$2 dollars for each hit. Let Y be his profit. Find the expectation and the variance of Y .
- (f) Now let's assume that the marksman enters the shooting range for free and gets the number of dollars that is equal to the square of the number of hits. Let Z be his profit. Find the expectation of Z .

3. 4 buses carrying 148 Tsinghua volunteers arrive at National Stadium. The buses carry 40, 33, 25, and 50 volunteers, respectively. One of the volunteers is randomly selected. Let X denote the number of volunteers that were on the bus carrying this randomly selected volunteer. One of the 4 bus drivers is also randomly selected. Let Y denote the number of volunteers on his bus.

- (a) Which of $\mathbf{E}[X]$ or $\mathbf{E}[Y]$ do you think is larger? Give your reasoning in words.
- (b) Compute $\mathbf{E}[X]$ and $\mathbf{E}[Y]$.

4. **St. Petersburg paradox.** You toss independently a fair coin and you count the number of tosses until the first tail appears. If this number is n , you receive 2^n dollars. What is the expected amount that you will receive? How much would you be willing to pay to play this game?

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- Random variables: A **function** from the sample space Ω to the real numbers
- Expectation

$$\mathbf{E}[X] = \sum_x x p_X(x)$$

- Variance

$$\text{var}(X) = \mathbf{E}[X^2] - (\mathbf{E}[X])^2$$

- Readings: Section 2.1-2.3, start 2.4
- Assignment: Problem Set 4
- Next lecture: Conditional expectation; Multiple discrete random variables